
Monocentric City Model II

LSE GY457 - AT 2025

Topic 5

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Course overview

- Block 1

- Intro to Urban and Regional Economics and course overview
- Topic 1: Regional and urban concentration forces
- Topic 2: The empirics of agglomeration
- Topic 3: The costs and benefits of agglomeration

- Block 2

- **Topic 4&5: Monocentric city model**
- Topic 6: Firm location choice

- Block 3

- Topic 7: Neighborhoods and segregation
- Topic 8: The vertical dimension of cities
- Topic 9: Gentrification and suburbanization
- Topic 10: Measuring cities with new technologies

Recap from last time

- Alonso-Mills-Muth (AMM) model is workhorse tool in urban economics
 - Assumes all employment is concentrated in the city center (CBD)
 - Spatial equilibrium implies lower rents at higher commuting distances
 - Housing rent decreases with distance to CBD
 - But: Housing consumption increases with distance to CBD
 - Developers respond to changes in housing rent
 - Land rent decreases in distance to the CBD
 - In real world, most cities tend to be centralized, but there are exceptions
-

Today: Monocentric city model II

- Monocentric city model II
 - Equilibrium conditions and additional restrictions
- Comparative statics: Income
 - City structure and income: Income segregation
- Comparative statics: Transportation costs
 - City structure and transportation costs: Decentralization
- Other predictions
 - Open city vs. closed city

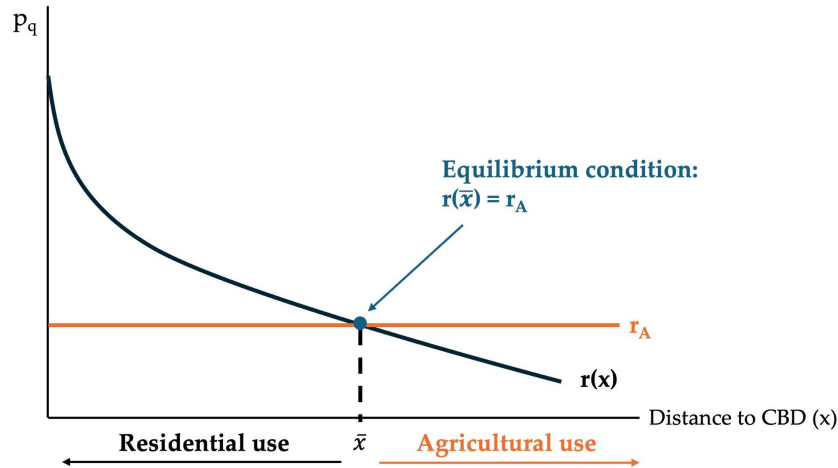
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First off: Where does the city end in this model?

- City is characterized by “urban” land
- Intuitively, for land to be urban, urban land rent must exceed agricultural land rent
 - Otherwise it would be more profitable for agricultural use
- Remember: in Alonso model, land rent is paid by households (who consume directly land) vs. in Muth-Mills model, land rent is paid by developers (that supply housing bought by households)

One more equilibrium condition



- At $\bar{x}$, the rent from agricultural land equals the rent from urban land
- More profitable to use land beyond $\bar{x}$ for agriculture, and land closer to $\bar{x}$ for residential use
- >>>> $\bar{x}$ defines the city border!
(endogenous point!)

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Some basics: Exogenous vs. endogenous variables

- *Exogenous variables* are determined outside of the model
 - Here: income (y), commuting costs (t), agricultural land rent (r_A)
- *Endogenous variables* are solved within the model
 - Here: land rent (r), housing rent (p_q), structural density (S), population density (D), city border ($\bar{x}$)
- In addition, in closed-city: Total population (N) is exogenous; utility ($\bar{U}$) is endogenous
 - Different from open-city case (as we will see later)

Some basics: Comparative static analysis

- Analyze the impact of a change in one exogenous variable on endogenous variables
- Allows us to use the structure of the model to establish counterfactuals and evaluate what would happen under alternative scenarios
- Derive predictions: What happens to rents, densities, etc. if income, transportation costs, or the opportunity cost of land change?
 - Relevant for policy makers, planners, developers, investors, etc.
 - Predictions differ depending on open vs. closed city (more later)

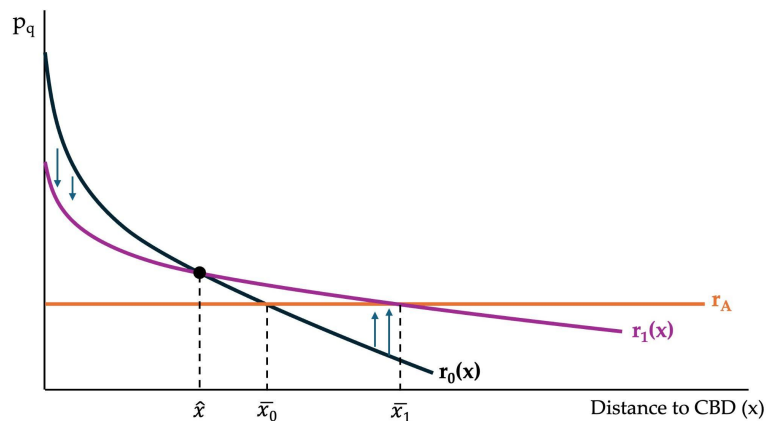
Comparative statics: Income $\uparrow$

- What happens to city structure (rent and population density gradients) if income increases?
- From Topic 4, the slope of the (housing) rent gradient is:

$$\partial p_q / \partial x = - t/q(y) < 0$$

- Where we make explicit the fact that the quantity of housing (q) depends on income (y)
- Both q and c are normal goods, meaning that if income increases, the household consumes more of both (i.e., $y \uparrow \Rightarrow q(y) \uparrow$ and $c \uparrow$)
- So that $y \uparrow \gg q(y) \uparrow \gg -t/q(y) \uparrow$, i.e. the slope of the rent gradient becomes less steep (it flattens)

Comparative statics: Income $\uparrow$



- As income increases, the bid-rent curve becomes flatter (from $r_0(x)$ to $r_1(x)$)
- Bid rents increase in the periphery, and more land is for residential use
- Population (fixed) moves to the suburbs
- Demand goes down in the center
- Bid-rent curve rotates around $\hat{x}$
- City expands (border moves from $\bar{x}_0$ to $\bar{x}_1$)

Comparative statics: Income $\uparrow$

- Housing rent decreases in the center but increases in the suburbs
 - Similarly, structural density, housing density, and land rent decrease in the center but increase in the suburbs
 - Housing consumption per capita increases in the center, as housing rent gets cheaper
 - In turn, population density decreases in the center
- Utility increases, as residents become richer and hence can consume more of everything

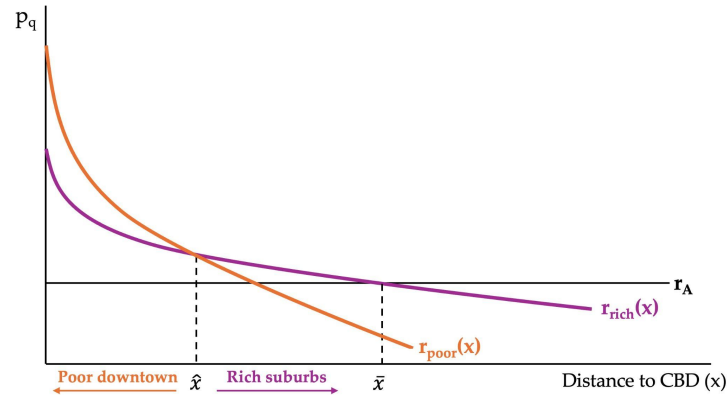
Comparative statics: Income $\uparrow$

- What happens to housing consumption?
 - Housing consumption per capita increases in the center, as housing rent gets cheaper
 - Housing consumption decreases in the suburbs, where housing rent gets more expensive
 - At the same time, in the new areas where the city has expanded (for $x > \bar{x}_0$), housing consumption must increase (as it was zero before)
 - Overall housing consumption for locations further than $\hat{x}$ is ambiguous

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Income segregation



- Think of the exact same scenario as if some residents get a positive income shock and others do not
- The bid-rent curve for rich households is flatter than that of poor households (as seen before)
- Prediction: Poor live in downtown, rich in suburbs
- This holds under the assumption that commuting costs are only monetary
 - i.e., no disutility of commuting

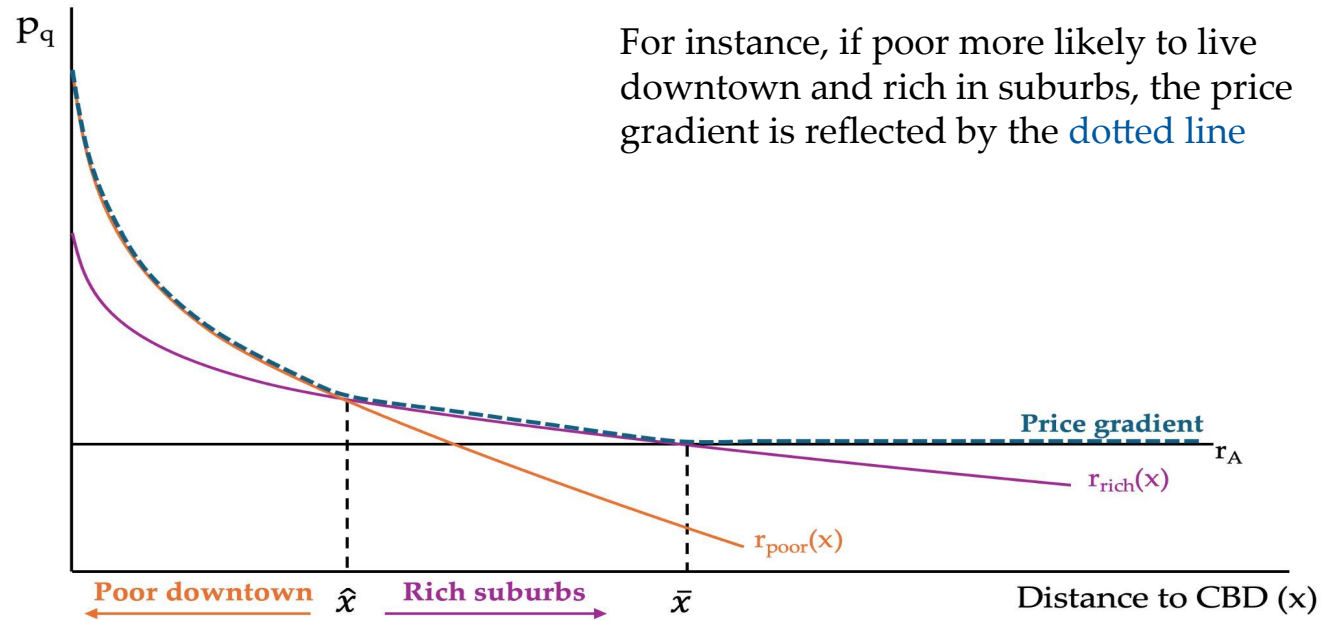
Income segregation

- With monetary commuting costs, rich live in the suburbs, poor live in downtown areas
 - Rich households spend more on everything, including commuting
- However, if we consider the value of time, commuting costs depend on income
 - Rich value time more (Why? e.g., as they earn more per hour)
- What happens to city structure? Results are ambiguous, as they depend on the desire for shorter commutes vs. preference for larger and cheaper houses outside center
 - Can explain differences across cities today (e.g. central Paris rich, downtown LA poor)

Income segregation: Bid-rent curves vs. price gradients

- Real cities populated by different types of households
- The observed price gradient within a city reflects a combination of different bid-rent curves
- The bid-rent curve of each type is convex (as we have seen in the generic case) as people tradeoff between space and distance (*substitution effect*)
 - Big houses and high commuting costs vs. small houses and low commuting costs
- The overall price gradient is even more convex because different types choose different locations (*sorting effect*)

Income segregation: Bid-rent curves vs. price gradients



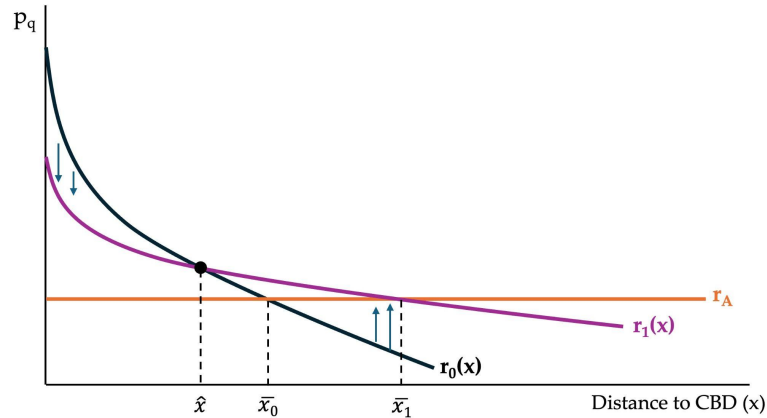
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Comparative statics: Transportation costs

- What happens to city structure (rent and population density gradients) if transportation costs decrease?
- We have seen that the housing rent gradient decreases with distance from CBD
 - From Topic 4: $\partial p_q / \partial x = -t/q < 0$
- If transportation costs decrease $t \downarrow \gggg -t/q(y) \uparrow$, i.e. the slope of the rent gradient becomes less steep (it flattens)
 - Similar to the effect of an increase in income

Comparative statics: Transportation costs



- As transportation cost decrease, the bid-rent curve becomes flatter
- Bid rents increase in the suburbs, and more land is for residential use
- Population (fixed) moves to suburban areas
- Demand for housing goes down in the center
- Bid-rent curve rotates around $\hat{x}$
- City expands

Comparative statics: Transportation costs

- Housing rent decreases in the center but increases in the suburbs
 - Similarly, structural density, housing density, and land rent decrease in the center but increase in the suburbs
- Housing consumption q inversely related to prices p
 - Housing consumption q increases in the center, where $p \downarrow$
 - q goes down in suburbs, where $p \uparrow$
- Population density $\downarrow$ in center; $\uparrow$ in suburbs
- Utility increases since the cost of commuting goes down
 - Similar to what happens with a positive income shock

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Decentralization

- We have been experiencing steady income growth in real terms over more than a century
 - Average global real GDP per capita growth between 1900 and 2016 equal to 1.72% annually (Maddison Project Data)
 - Could lead to decentralization
- Major transportation improvements since late 19th century
 - Subway and commuter rail; cars and highways
 - Make travels easier and cheaper
 - Should lead to decentralization
 - *What is the empirical evidence?*

Causes of decentralization

- There is strong evidence of urban decentralization
 - In line with transportation improvements since the mid 19th century (rail and road networks)
- But: Other explanations are also possible
 - Increase in income (AMM with monetary commuting costs)
 - Increase in preference for suburban areas
 - Increase in downtown disamenities (pollution, noise, crowdedness)
 - Aging of downtown housing; Other?
- Is there *casual* evidence of the impact of improvements in transportation infrastructure on decentralization?

Baum-Snow (2007): “Did highways cause suburbanization?”

TABLE I
AGGREGATE TRENDS IN SUBURBANIZATION, 1950–1990

	1950	1960	1970	1980	1990	Percent change 1950–1990
Panel A: Large MSAs						
MSA population	92.9	115.8	134.0	144.8	159.8	72
Total CC population	44.7	48.5	51.3	49.2	51.0	14
Constant geography CC population	44.7	44.2	42.6	37.9	37.1	-17
N for constant geog. CC population	139	132	139	139	139	
Panel B: Large Inland MSAs						
MSA population	39.2	48.9	57.0	65.0	73.5	88
Total CC population	16.8	19.7	22.1	22.1	23.2	38
Constant geography CC population	16.8	16.5	15.4	13.3	12.5	-26
N for constant geog. CC population	100	94	100	100	100	
Total U. S. population	150.7	178.5	202.1	225.2	247.1	64

Estimate how much of this change can be attributed to highway construction

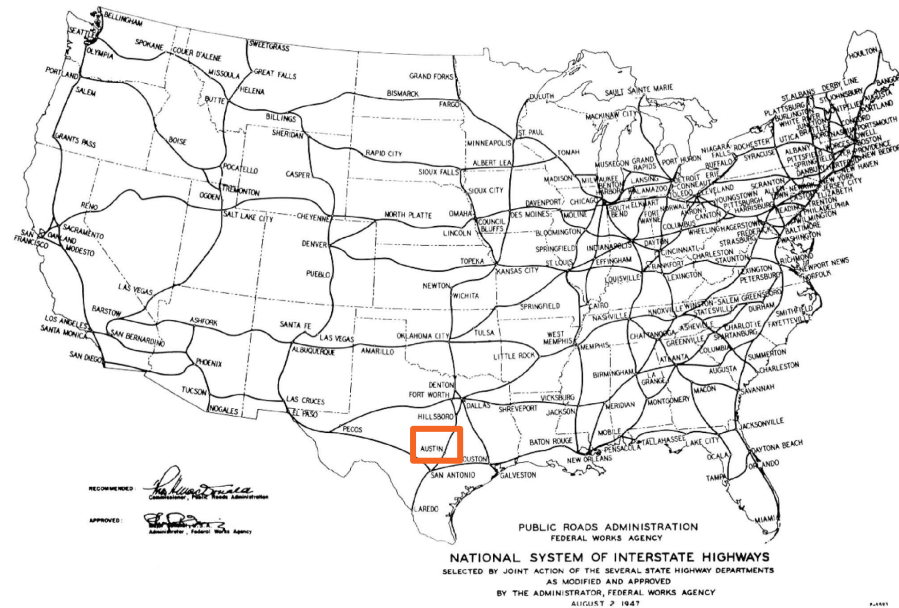
Baum-Snow (2007): “Did highways cause suburbanization?”

- Chicken-and-egg question: Did people move to the suburbs *because* of the highways or were highways built in cities that were already suburbanizing?
 - Are highways causing decentralization or is decentralization favoring the construction of highways (as there is increase demand for this type of infrastructure)?
 - Highway assignment can be endogenous to local economic conditions >>>> Endogeneity issue!
- Estimation method: Instrumental variables
 - IV: number of rays predicted by *national* plan of highways

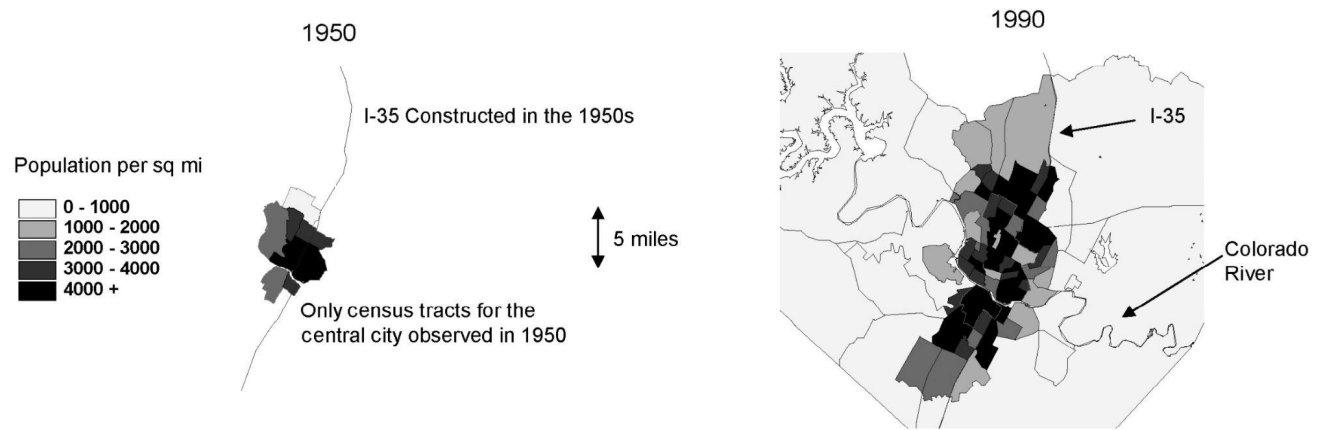
Baum-Snow (2007): “Did highways cause suburbanization?”

- Idea: US Interstate Highway System was designed to link far way locations and not to facilitate local commuting within a metropolitan area
 - 1947 Highway plan meant to facilitate trade and national defense
 - Some metro areas receive more highways because they are nearer to other (relevant) population centers
- Empirical strategy: Compare metro areas that received many highways (because of *exogenous* reasons) between 1950-1990 to those receiving fewer highways

Baum-Snow (2007): “Did highways cause suburbanization?”



Baum-Snow (2007): “Did highways cause suburbanization?”

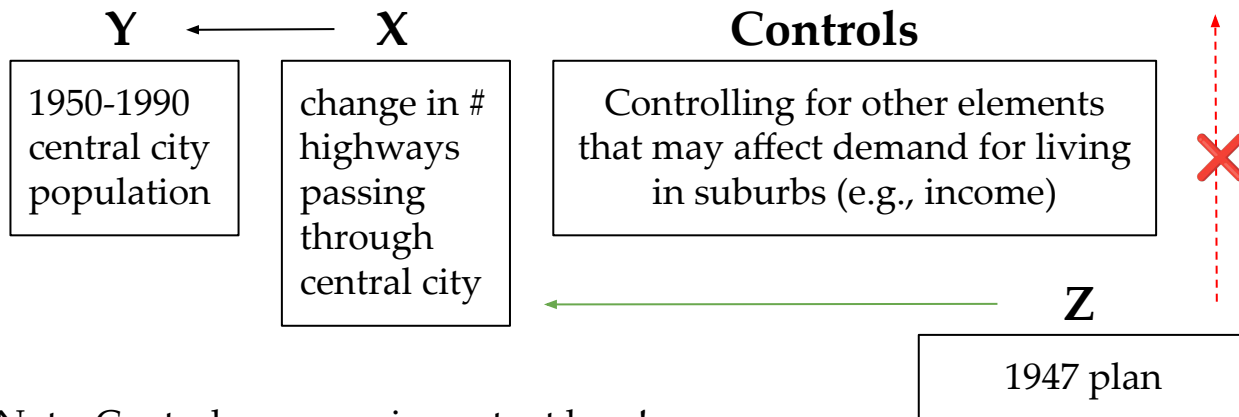


Note: Each shaded region is a separate census tract.

FIGURE II
Development Patterns in **Austin, TX.**

Baum-Snow (2007): “Did highways cause suburbanization?”

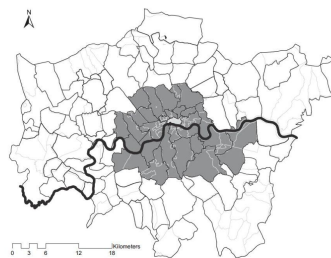
$$\Delta \log N_i^c = \delta_0 + \delta_1 \Delta \text{ray}_i + \delta_2 r_{ci} + \delta_3 \Delta \tilde{w}_i + \delta_4 \Delta \log N_i^{\text{MSA}} + \delta_5 \Delta \tilde{G}_i + \varepsilon_i$$



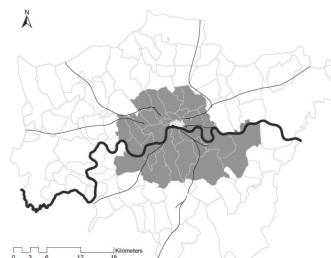
Note: Controls are very important here!

Main result: *one* new highway passing through a central city reduces city population by around 18%

London and “The making of the modern metropolis” (Heblich et al., 2020)



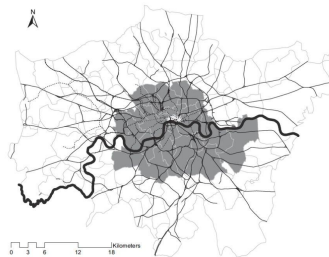
(A) Administrative Boundaries



(B) Railway Network 1841



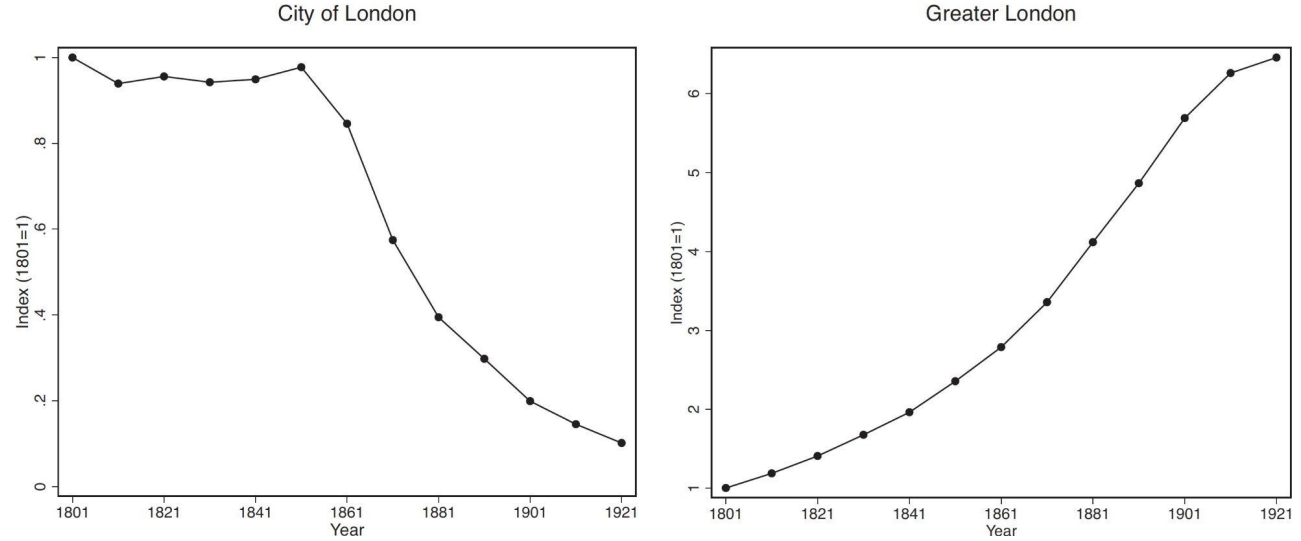
(C) Railway Network 1881



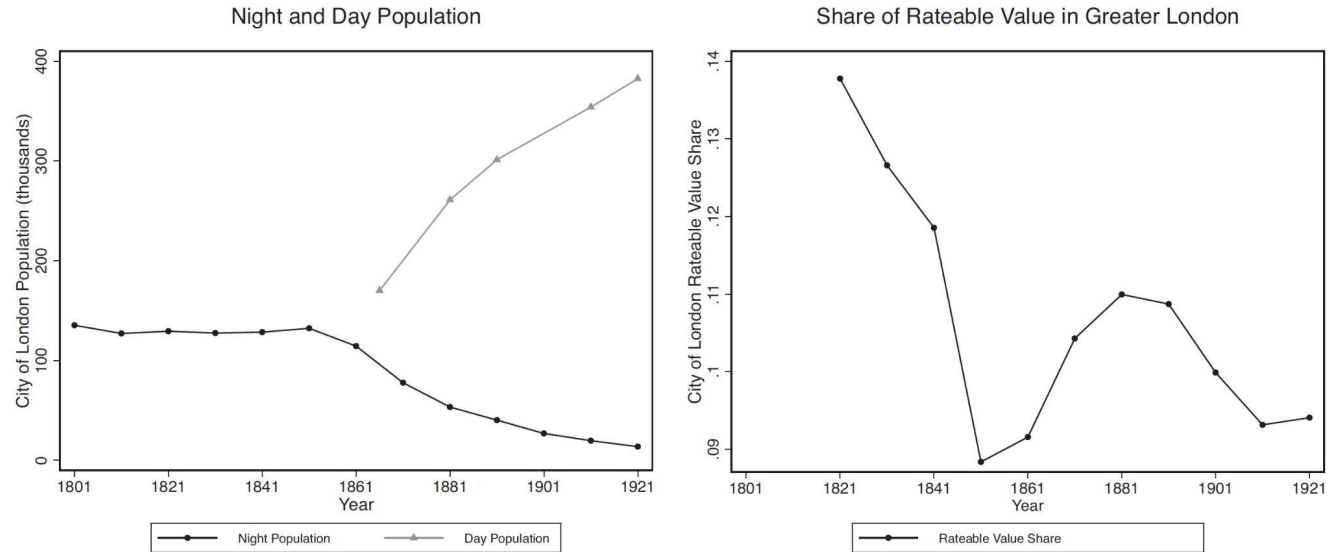
(D) Railway Network 1921

- Show that the invention of steam railway in London led to the first large-scale separation of residence and workplace locations
- Using (a lot!) of algebra, they generate a rich spatial model
- Removing railway network, population and land value ↓50%

London and “The making of the modern metropolis” (Heblich et al., 2020)

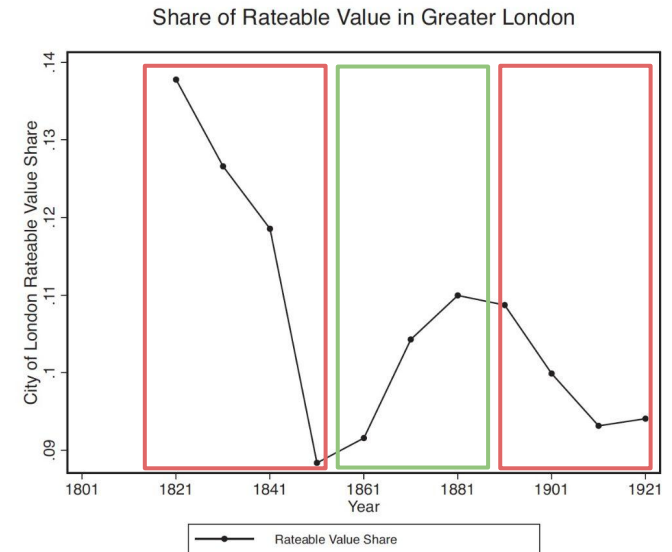
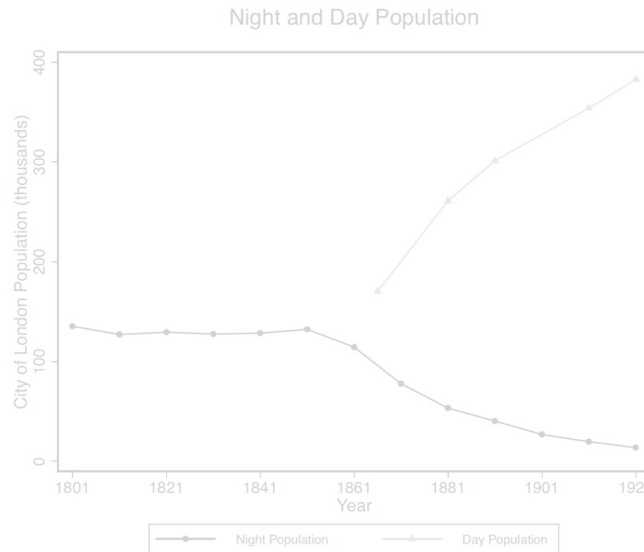


London and “The making of the modern metropolis” (Heblich et al., 2020)



London and “The making of the modern metropolis” (Heblich et al., 2020)

What is happening here?



London and “The making of the modern metropolis” (Heblich et al., 2020)

To examine the implications of this change in specialization for the value of land and buildings, the right panel of [Figure III](#) displays the City of London’s share of Greater London’s rateable value over time. In the early nineteenth century, this share declines from around 14% to 9%, which is consistent with a geographical expansion in the built-up area of Greater London. As this expansion occurs, and undeveloped land becomes developed, the share of already-developed land in overall land values tends to fall, because the denominator of this share increases. In contrast to this pattern, in the years after 1851 when the City of London experiences the largest declines in its residential population, its rateable value share increases from 9% to 11%. Finally, in the decades at the end of the nineteenth century, the pattern of a decline in this rateable value share again reasserts itself, consistent with the continuing geographical expansion in the built-up area of Greater London. Therefore, the change in the City of London’s pattern of specialization in the decades immediately after 1851 increases the relative value of this location.

(p. 2078)

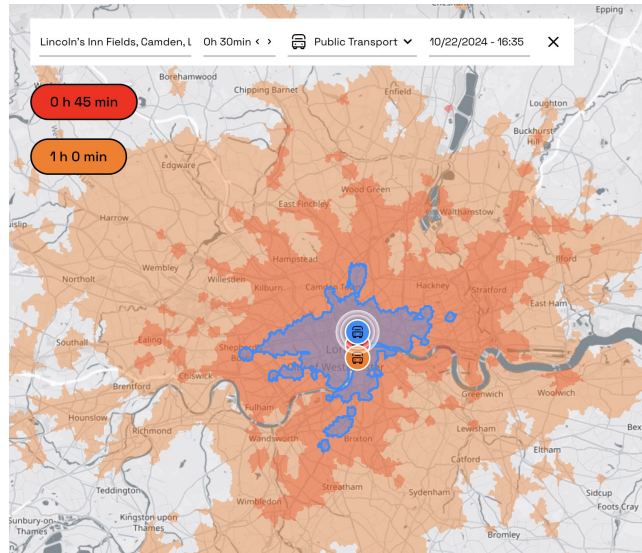
Urban decentralization

- Cities around the world have been decentralizing in terms of prices (housing rent, land price) and quantities (densities)
- Causal evidence shows that part of this trend is due to better transportation
 - If we build radial connections, urban structure will decentralize
 - Implications: People move to suburbs and travel longer distances; demand and prices for housing increase in suburbs
- Important for predictions: Relevant for planners, developers, investors, etc.

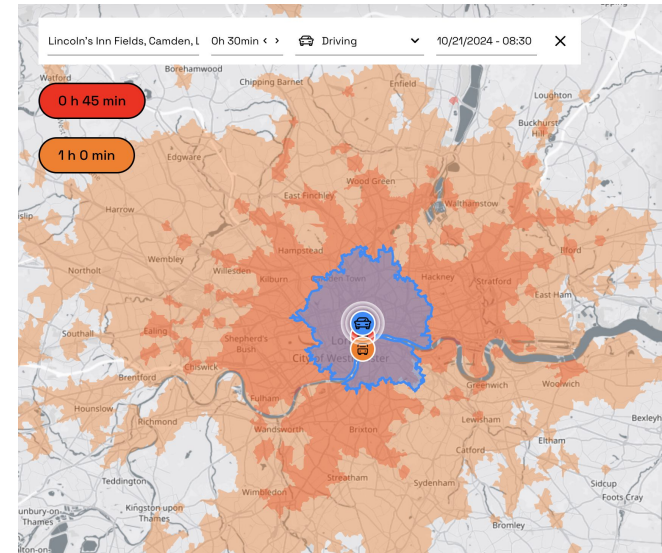
London: Commuting times today

Locations accessible within 30/45/60 min from Lincoln's Inn Fields

Public transport



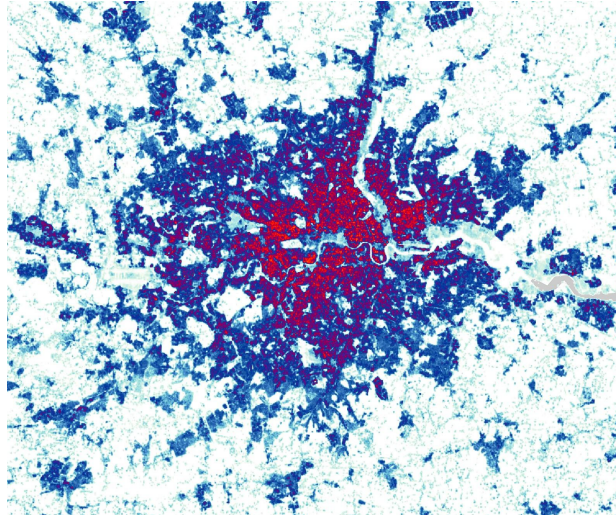
Private transit



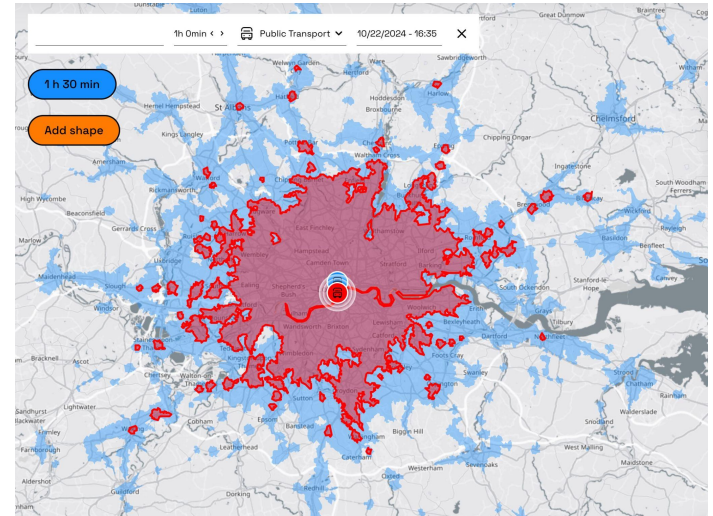
Source: <https://app.traveltime.com/>

London: Pop density and commuting

Population density

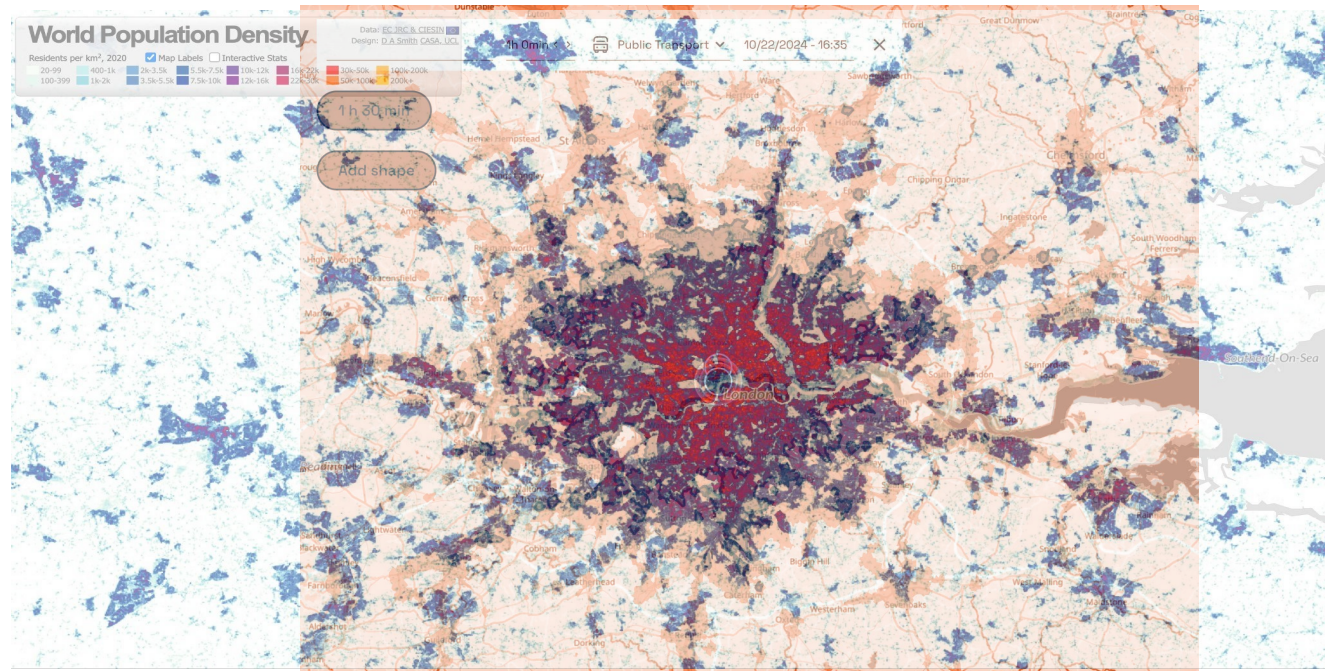


Locations accessible within 60/90 min from center



Sources: <https://luminocity3d.org/WorldPopDen/#3/20.00/10.00>; <https://app.traveltime.com/>

Let's overlay them!



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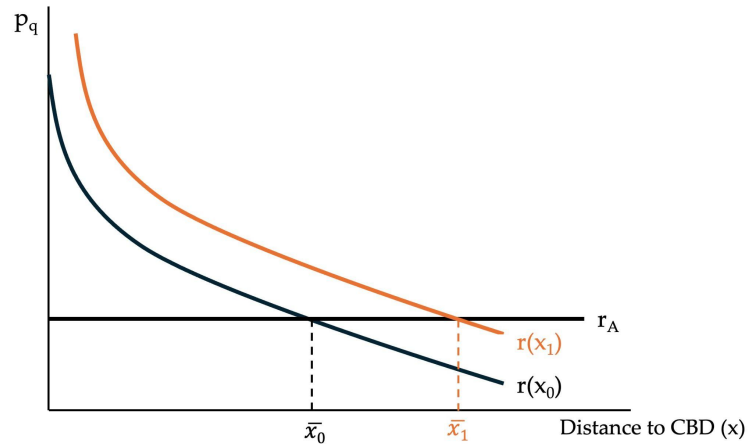
What are other (interesting) comparative statics we can look at?

- Remember the variables we can “shock” are the ones exogenous to the model
 - In the closed-city case, they are: income (y), commuting costs (t), agricultural land rent (r_A), and total city population (N)
 - We have already seen what happens if y , t change
 - We can now see what happens to the equilibrium if $N \uparrow \downarrow$ or $r_A \uparrow \downarrow$

Comparative statics: Total population $\uparrow$

- What happens if $N \uparrow$?
 - Draw the graph
 - Think about what happens to prices, housing consumption, population density, structural and housing densities, and in turn to the city overall; also, what happens to utility?

Comparative statics: Total population $\uparrow$

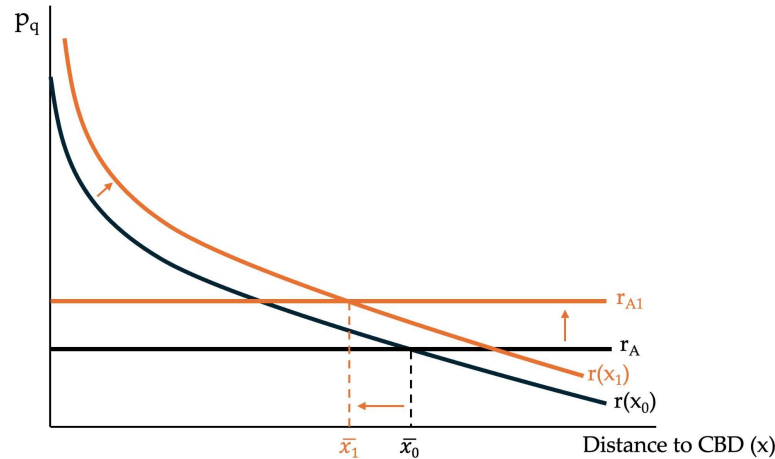


- Increase in population N increases demand in all locations in the city
- Prices increase (both land rent and housing rent)
- Housing consumption decrease
- Structural/housing density, and population density increase
- City expands
- Utility decreases

Comparative statics: Agricultural land rent↑

- What happens if $r_A \uparrow$?
 - Draw the graph
 - Think about what happens to prices, housing consumption, population density, structural and housing densities, and in turn to the city overall; also, what happens to utility?

Comparative statics: Agricultural land rent↑



- If $r_A \uparrow$, city shrinks
- Developers can make higher profits converting some urban land into agricultural land
- Some people relocate
- Prices increase
- Housing consumption↓
- Structural/housing density↑, pop density↑
- Utility↓ (ignoring agglomeration benefits)

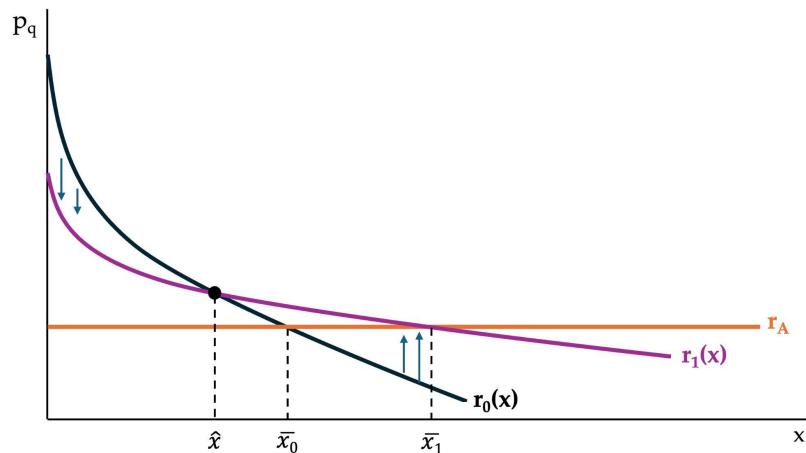
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Open-city case

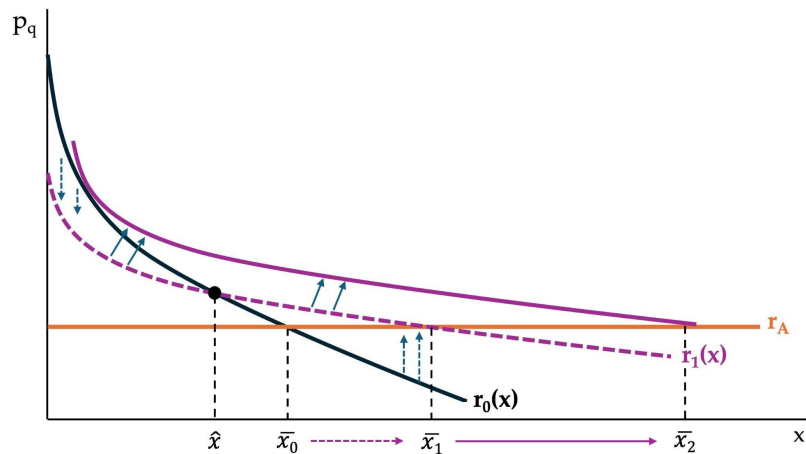
- vs. Closed city: People can move in and out of the city
 - Total city population N is endogenous
 - City-level utility $\bar{U}$ is fixed (exogenous)
 - Mechanics: People move until reach same utility (equilibrium)
- Predictions are similar to the closed-city, but algebra to get there is a bit more complex
 - Idea: Hold N constant and evaluate the changes in city structure using the closed-city model (as done before)
 - Adjust population N to offset utility differences (since people are free to move across cities, they will move to the city offering higher utility until the utility is equalized again across cities)
 - If $N \uparrow$, demand for housing $\uparrow$, prices $\uparrow$, housing consumption $\downarrow$

Open-city case: Income $\uparrow$



- Remember what happens in the [closed city](#)
- Bid-rent becomes flatter: demand goes down in center, goes up in suburbs
- Population moves to suburbs, city expands, and central population $\downarrow$

Open-city case: Income $\uparrow$



- In open city, an increase in income makes the city more attractive to people *outside* ($y \uparrow \gg U \uparrow$)
- Migration amplifies the effects on prices and density...
- ...and expansion to suburbs does not imply population loss in the center!

Summary of predictions (Brueckner, 1987)

CLOSED CITY	Effect on	Housing rent		Housing consumption		Structural density		Population density		Land rent		U
	City area	City	Sub	City	Sub	City	Sub	City	Sub	City	Sub	
Income ↑	↑	↓	↑	↑	↔	↓	↑	↓	↑	↓	↑	↑
Commuting costs ↑	↓	↑	↓	↓	↑	↑	↓	↑	↓	↑	↓	↓
Agricultural land rent↑	↓	↑	↑	↓	↓	↑	↑	↑	↑	↑	↑	↓
Population ↑	↑	↑	↑	↓	↓	↑	↑	↑	↑	↑	↑	↓
OPEN CITY	Effect on	Housing rent		Housing consumption		Structural density		Population density		Land rent		Pop
	City area	City	Sub	City	Sub	City	Sub	City	Sub	City	Sub	
Income ↑	↑	↑	↑	↓	↓	↑	↑	↑	↑	↑	↑	↑
Commuting costs ↑	↓	↓	↓	↑	↑	↓	↓	↓	↓	↓	↓	↓
Agricultural land rent↑	↓	—	—	—	—	—	—	—	—	—	—	↓

When should we use open vs. closed city model?

- Closed city: Assumes no cross-city migration
- Open city: Free cross-city migration
- May be worth asking: Do we expect cross-city migration?
 - If all cities change the same way, no migration incentives
 - If one city expected to change, migration incentives (in or out)
- Use closed-city model to predict changes occurring in *all cities* in a country/region
- vs. use open-city model to predict changes occurring in *one city* in a country/region

Summary

- Monocentric city model delivers useful predictions
 - Higher income leads to decentralization (when considering monetary commuting costs)
 - Better transportation infrastructure leads to decentralization
 - Higher agricultural land rent shrinks cities
- Some predictions validated with causal evidence
 - Cities decentralize with better transportation infrastructure
- Predictions useful for planners, developers, investors
 - Give insights into expected effects on demand, prices, densities

Next time

- Firm location choice
 - Firms in the monocentric city model
- Agglomeration and decentralization
 - Endogenous agglomeration
 - Multiple equilibria
- Emergence of new clusters
 - Sub-centers
 - Edge cities
 - (Historic) anchoring

Thank you!

Readings and references

Required readings

- Baum-Snow, N. (2007). Did Highways Cause Suburbanization? Quarterly Journal of Economics, 122(2): 775-805.
- DiPasquale and Wheaton, Ch. 3, The Urban Land Market: Rents and Prices.
- McDonald and McMillen, Ch. 6, Introduction to urban location patterns: Static analysis; Ch.7, Using the Monocentric City Model.

Other readings and references

- Brueckner J. (1987). The Structure of Urban Equilibria. In Handbook of Regional and Urban Economics, Vol II, 821-846.
- Heblich, S., Redding, S. J., and Sturm, D. M. (2020). The Making of the Modern Metropolis: Evidence from London. Quarterly Journal of Economics, 135(4): 2059-2133.